TAKUDZWA WILSON MUZAVAZI

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PROFESSIONAL SUMMARY

Geospatial professional with 3 years of experience in infrastructure mapping, environmental monitoring, and ecological research across Zimbabwe and the United States. Combines field data collection, remote sensing, and spatial analysis to turn geospatial data into maps and insights that support planning, risk assessment, and research. Brings a strong academic foundation in geographic information science and earth observation, complemented by campus leadership and active involvement in the GIS community.

TECHNICAL SKILLS

GIS Software & Programming: ArcGIS Pro, QGIS, ArcGIS ModelBuilder, Python, R

Spatial Analysis, Remote Sensing & Cartography: Geoprocessing, raster and vector analysis, hotspot analysis, satellite imagery processing, land-use change analysis, thematic mapping

Geospatial Data Management & Field Collection: Geodatabase management, data validation, georeferencing, digitization, GPS, ArcGIS Field Maps, Open Data Kit (ODK)

PROFESSIONAL EXPERIENCE

MICHIGAN TECHNOLOGICAL UNIVERSITY — **Houghton, Michigan**

Research Assistant — *August 2025 - Present*

- Analyze the spatial distribution of moose skeletal remains using ArcGIS Pro to investigate ecological patterns and habitat-related mortality.
- Maintain a geodatabase of 5000+ moose records, integrating skeletal records, GPS locations, and ecological attributes.
- Create maps and spatial visualizations using ArcGIS Pro and R to communicate wildlife distribution patterns and support ecological research.

MAP TECH PVT LTD T/A ZIMGEO — **Harare, Zimbabwe**

GIS Assistant — *May 2024 – May 2025*

Conducted GIS analysis of agricultural land use and mining environmental data to support environmental monitoring and planning.

- Georeferenced and digitized historical aerial photographs in QGIS for integration with modern spatial datasets.
- Processed Landsat and Sentinel-2 imagery in QGIS to analyze land-use change and environmental trends, supporting environmental planning and decision-making.

GLENRISE — **Ngezi, Zimbabwe**

Environmental Assistant — *February 2022 – September 2022*

- Analyzed 125 mining incident records in QGIS using buffering and kernel density analysis to identify operational risk zones and support mine safety planning.
- Prepared 13+ environmental monitoring and compliance reports by combining geospatial data, field observations, and statistical summaries.
- Produced 10 thematic maps and 13 spatial reports to communicate environmental findings to project stakeholders.

NATIONAL RAILWAYS OF ZIMBABWE — **Harare, Zimbabwe**

GIS Intern — *June 2018 – July 2019*

- Collected and validated railway infrastructure and derailment-location data using GPS and satellite imagery, integrating information into GIS layers for analysis.
- Applied hotspot and spatial pattern analysis in QGIS to identify derailment risk zones and support targeted maintenance planning.
- Digitized and mapped 300 km of railway infrastructure for network analysis, infrastructure planning, and operational risk assessment.

EDUCATION

MICHIGAN TECHNOLOGICAL UNIVERSITY — **Houghton, Mi**

Master of Science in Geographic Information Science — April 2027

- Relevant Coursework: Data Science for Environmental Sensing, Python for Modern GIS, Applied Spatial Statistics, Advanced GIS Concepts & Analysis, Remote Sensing of the Environment

UNIVERSITY OF ZIMBABWE — **Harare, Zimbabwe**

Bachelor of Science in Geographic Information Science and Earth Observation — December 2020

- Thesis title: Detecting Glycaspis Brimblecombei using Sentinel 2 multispectral imagery.

PROJECTS

Asset Mapping & Data Validation — ArcGIS Field Maps, GPS, Geodatabases and GIS digitization.

Verified Upper Daniel Heights asset locations against drone imagery. Digitized railway assets and mapped derailment hot spots for the National Railways of Zimbabwe.

Spatial Analysis & Remote Sensing — Proximity analysis, IDW, and kernel density

Analyzed moose distribution by shoreline distance on Isle Royale National Park and mapped vegetation change in Calumet using MODIS imagery and NDVI.

Python Geospatial Modeling — ArcGIS Pro and spatial interpolation

Compared two Empirical Bayesian Kriging methods for snow depth using cross-validation in an ArcGIS Pro Python notebook.

LEADERSHIP AND VOLUNTEER EXPERIENCE

GIS Student Volunteer **Milwaukee, Wisconsin**

GIS-Pro 2026 *October 2026*

- Supported workshop delivery, hands-on activities, and certification administration for the Cartography and Map Design Boot Camp: Theory and Best Practice Fundamentals.

Secretary **Michigan Technological University**

African Student Organization (ASO) *April 2026 – Present*

- Assist in planning and execution of cultural events and community outreach programs promoting African heritage on campus.

Geospatial Professional Network (GPN) Member *April 2026 – Present*

American Association of Geographers (AAG) Member *April 2026 – Present*